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CHANGE HISTORY

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DOCUMENT REVIEW AND APPROVAL

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		Document Definitions	
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TABLE OF CONTENTS

Purpose	4
Definitions	4
Document Definitions	4
Table 01-1 Document Definitions	4
Document Acronyms	5
Table 01-2 Document Acronyms	5
Customer Needs	6
Problem Statement.....	6
Customer Needs Description	6
Key Stakeholders	7
General Constraints	7
External Factors - Global, Cultural, and Environmental	7
Social Factors - Public Health, Safety, and Welfare	8
Ethical Factors - Global, Societal, Economic, and Environmental	9
Use Case	9
Product Requirements	11
System Requirements	11
Target Specifications	20
Table 01-5 Target Specifications.....	21
Table 01-6 Standards and Statutory Requirements.....	23
Benchmarking Information	24
Table 01-7 Benchmarking Information	24
References	25

Purpose

This document contains the customer needs and design requirements (functional and engineering metrics) with target values for the Resilient Mesh Network Testbed for Mobile/Ad-Hoc Nodes. This is a ground-based, UAV-inspired wireless mesh testbed sponsored by the Idaho National Laboratory (INL). It is intended to establish and communicate the customer's needs, as captured in INL Project Request Form S27-48 [1] and the 15 September 2026 customer meeting [2]. This document will be used as entry criteria to the Architecture Phase of the project (fall semester).

Definitions

Document Definitions

Table 01-1 Document Definitions

Term	Definition
Design History File	A compilation of records containing the complete design history of a finished device or service
Verification	Confirmation by testing, analysis, demonstration, and/or inspection that specified requirements have been fulfilled
Validation	Establishing objective evidence that system specifications conform to user needs and intended uses in the operational environment
Component	One of the parts that make up a system. A component may be hardware or software and may be subdivided into components
Functional Testing	Testing that ignores the internal mechanism of a system or component and focuses on the outputs generated in response to selected inputs
Mobile Ad-hoc Network	A self-configuring wireless network of mobile nodes that operates as both hosts and routers
Virtual node	A software-emulated node that runs the same routing protocol as the physical nodes. They join the same mesh to reach swarm-scale node counts.

Mesh Node	The physical, battery-powered, Raspberry Pi-class board computer with a wireless interface used to participate in the mesh
Self-healing	The ability of the network to detect node/link loss and restore connectivity by rerouting traffic without user interference.
Reconvergence	The process where a computer network recalculates and updates its routing paths after a change in topology.
Multi-hop	A communication method where data packets travel through one or more intermediate nodes to get from a source destination.
Link quality	Measures how reliably and accurately data packets travel between a transmitter and a receiver
Packet delivery ratio	The percentage of data packets successfully received at their destination compared to the total number of packets sent by the source
Topology	The physical and logical arrangement of devices and connections in a computer network

Document Acronyms

Table 01-2 Document Acronyms

Acronym	Description
DHF	Design History File
IEEE	Institute of Electrical and Electronics Engineers
CR	Customer Requirement
SR	System Requirement
INL	Idaho National Laboratory
FCC	Federal Communications Commission
EHS	Environmental Health and Safety

RF	Radio Frequency
GHz	Gigahertz
Mbps	Megabits per second
BATMAN	Better Approach To Mobile Ad-hoc Networking

Customer Needs

Problem Statement

Swarms of UAVs supporting disaster response and infrastructure inspection depend on mobile ad-hoc networks in which nodes join, move, and disconnect without warning. These abrupt shutdowns are caused by power failure, jamming, or physical loss [7]. Researchers at Idaho National Laboratory studying distributed control strategies for wireless networks need lab-representative data on how quickly the mesh networks can detect these events and reroute the rest of the existing nodes. However, flying a swarm of ten or more drones is costly and almost impractical for implementation [7].

Over the next couple months, our project group is tasked with designing and building a ground-based, scale-model mesh network testbed in Blacksburg. Four to six Raspberry Pi-based mobile nodes, supplemented by virtual nodes, will simulate a self-healing ad-hoc routing protocol using mobile robots provided by the university. Several tests being run will include physically moving the nodes and intentionally powering them down to evaluate reconvergence of time, throughput, and latency. The feedback we receive from multiple runs will assist in improving scalability. Due to existing implementations of the mesh testbed, our team's priority will be searching for a repeatable way to measure swarm-network resilience.

Customer Needs Description

INL needs a physical testbed that reproduces the connectivity challenges of a UAV swarm. This involves moving nodes, links degrading, and nodes disappearing without any alert to the user. However, our implementation will be ground-based and produce quantified resilience data. The customer emphasized that absolute speed matters less than how fast nodes join and how the network recovers. Our baseline scalability should be beyond 4-6 nodes, potentially up to 10+ nodes in later iterations. The preferred configuration is four physically moving nodes combined with about six simulated nodes [1].

The customer also requested the focus to be on routing protocols rather than hardware, learning more towards Layer 2 mesh protocols such as BATMAN. Additionally, we must carry a representative data stream such as standard-quality video at about 2 Mbps. Telemetry must

have enough time resolution to show exactly when and where the nodes break, and we're expected to actively seek out the conditions that break the protocol while excluding systemic errors. A live visualization, such as a map of node positions with link quality is required. An acceptable approach to this would be a local Python application with screen recording.

Practical buildability within the course schedule is a core responsibility. This includes low-cost mobile platforms (robotics with Raspberry Pi's attached), swappable batteries giving 15-20 minutes charge, node spacings up to 25m, and rigorous removal of unnecessary hardware. Encryption and other tools for security are considered our stretch goals, potentially revisited in the following semester when physical implementations begin.

Key Stakeholders

The following key stakeholders are affected by the testbed:

1. INL Sponsor and Research Team (external): April Augustine and Nicholas Kaminski define the specifications and metrics and will use our provided resilience data to inform INL research on distributed control of wireless networks. They are our primary customers and approve of the following requirements.
2. SP27-48 MDE team and Mentor Joe Adams (internal): The team designs, builds, and tests the system and is evaluated on the deliverables given by INL. Our mentor oversees our implementation and enforces VT policies on budget, travel, safety, and any particular restrictions.
3. Future research teams (internal/external): The primary customers expect later iterations to grow the testbed into a 10+ node swarm inspired by existing UAVs. Therefore, future MDE teams as well as other sponsors are directly affected by our documentation.
4. Communities (external): Emergency responders, infrastructure operators, and public communities near UAV operations benefit from the resilient swarm communications. Our reports will follow IEEE standards to provide safety and privacy implications for future implementations.

General Constraints

External Factors - Global, Cultural, and Environmental

The testbed must validate mesh network performance against universal standards to ensure protocols and hardware are deployable globally. This requires standardized 2.4 GHz spectrum sharing hardware and BATMAN routing protocol implantation [1], enabling future adaptation to cover disaster scenarios and geographics regions. Documentation must be easy to interpret between differing cultures for universal understanding to create a sustainable future for the product created.

The testbed design prioritizes open-source tools and modular architecture to enable adoption by international research teams and future deployments in resource-constrained settings [1]. Documentation will accommodate both centralized research environments and decentralized field teams, supporting diverse emergency response workflows across different regions. This hardware also must be mass producible like Raspberry Pi's and accessible pricewise for countries to invest in [10].

The swarms will be deployed in environmental disaster zones ranging from hurricanes to sandstorms, as well as different mixtures of architecture that contribute to the environment requiring reliable link delivery probability [1]. Our testbed will need to simulate and test these environments accurately, measuring the link delivery probability and RF propagation characteristics, since we will be unable to place them in these situations realistically. This can be done through environmental testing using spectrum analysis for measurement and emulation of harsh channel conditions. Site-specific signal strength measurements can be characterized under typical harsh environments using spectrum analysis tools [11]. We must characterize link stability and path loss under disaster-representative conditions (temperature ranges 0-50° C, humidity 60-95%, multipath fading from mixed architecture) [1] and validate that link delivery probability ratio remains >80% and transmission lengths up to 25m acceptable under these conditions.

Social Factors - Public Health, Safety, and Welfare

The resilient mesh network testbed directly impacts the public health and welfare of emergency responders responsible for the safety of communities in disaster scenarios. When hurricanes, forest fires, or infrastructure failures disrupt communication networks, these responders depend on the self-healing ad-hoc networks to coordinate rescue operations, locate survivors, and avoid potential growing dangers. Our testbed design prioritizes reliability and accessibility to ensure first responders under high pressure scenarios can effectively deploy and trust these devices in critical scenarios. By ensuring our design has quick reconvergence, multi-hop redundancy, and operation under harsh environmental conditions, our product will be evidential in reliability for success in deployment zones.

For everyday testing, our main site will be Virginia Tech's main campus, which could have effects on students with RF exposure through 2.4 GHz and fire or thermal runaway from dealing with lithium batteries. To combat these possible scenarios, we will operate all transmitters within FCC 47 CFR Part 15 limits [3] and ensure all lithium batteries to be charged and stored according to Virginia Tech EHS guidelines [4]. There will also need to be harsh environment simulating where at Virginia Tech we have laboratories such as the Center for Packaging and Unit Load Design [12] where we can test for differing ranges of humidity and temperatures representing a vast range of global environments. We will also be testing under environmental factors such as wildland fire and high wind scenarios that may come from hurricanes, typhoons, or tornadoes at the Extreme Environments and Materials Lab [13]. When testing in these labs,

we ensure that the product being released is something operators can rely on to be accurate and withstand varying environments to successfully restore networks in times of peril.

When it comes to the safety standards in engineering design, we need to ensure that network reliability is prioritized and there are available diagnostics to support responder decision-making in disaster environments. Reconvergence time targets (PER-4: $\leq 2.0s$) and packet delivery baselines (PER-5 $\geq 95\%$) align with proper reliability standards and establish performance thresholds responders can trust. FUN-12 ensures that system monitoring survives network partition, allowing responders to distinguish real protocol failures from hardware faults (FUN-13), preventing false confidence in field conditions.

Ethical Factors - Global, Societal, Economic, and Environmental

The goal of our product is to create a secure, restorative self-healing mesh network that helps communities in times of disaster or fallen infrastructure. This will bring responders an easier job without potential hiccups of communication, making the mission more effective. A major IEEE Code of Ethics principle this project follows is I.1, mentioning the need “to hold paramount the safety, health, and welfare of the public” [14]. Another code is II.7 discussing the need to treat all persons fairly and with respect, which is related because of how this product will be used to help serve across the globe despite different incomes and infrastructure status.

When looking at it in a global context, our design takes an open source, modular design that provides future improvements and adaptation for the world. Societally, we take a transparent approach having public dissemination and open data creating trust by our users as well as the ability for our creation to be built upon between shared knowledge of communities.

Economically, the main hardware component being a Raspberry Pi leads to a cheaper design which was always the goal for all countries around the world to be able to source our design. Environmentally, we ensure our product is capable of being able to serve communities in any scenario that it may come across globally when infrastructure falls.

When creating a product that is meant to be a critical tool in disaster response scenarios, our team ensures that we are shipping something that is thoroughly tested and safe. We will be doing rigorous testing of our device, making sure that it can withstand varying temperatures and humidities (PER-3) [1], as well as environmental disasters such as wildfires or high-wind events [1]. We also will ensure that reliability in node-to-node connectivity through proper packet delivery ratio (PER-5: $\geq 95\%$).

Use Case

1. **Title**
 - Mesh networks for restoring communications in disaster zones.
2. **Primary Goal**

- Establish a self-healing ad-hoc communication network to coordinate rescue operations when infrastructure is damaged or unavailable.
- 3. Actors**
 - The main initiator that leads deployment of mesh networks.
 - First responders on scene for disaster zones, local survivors, remote operators monitoring mesh status.
 - 4. Preconditions**
 - Mesh nodes that are available and charged up.
 - Teams that have proper training for deployment.
 - A form of disaster leading to centralized network infrastructure going down.
 - 5. Trigger**
 - Sudden loss of communication between dispatchers
 - On-scene responders report issues or unavailability with communication tools.
 - Operators confirm the reason for this issue and is flagged.
 - 6. Main Success Scenario (Normal Flow)**
 - Once the main initiator receives a call for down infrastructure, there are teams sent to deploy 4-6 mesh nodes across the area.
 - Nodes form multi-hop ad-hoc mesh without centralized access point.
 - The network reaches a stable state within 2 seconds.
 - Responders can communicate across nodes with > 95% message delivery reliability.
 - Rescue operations coordinate via mesh, and throughput can support communication.
 - 7. Alternative Flows**
 - If nodes were placed in poor placement for efficiency the network will reroute around them (FUN-4)
 - If nodes have partial power loss or degrading issues, they will be able to reroute from said nodes (FUN-4)
 - Nodes are moved around by responders and able to keep stable connection (FUN-9)
 - 8. Exception Flows (Error Conditions)**
 - Nodes were never properly charged or stored properly by personnel (FUN-4)
 - Hardware fault occurs (Raspberry Pi crash, battery failure) which can be analyzed with system diagnostics that distinguish hardware from protocol failures (FUN-13 or POW-3)
 - 9. Postconditions (Outcomes)**
 - Network remains operational and supports ongoing rescue coordination.
 - Responders can continue working without communication issues.
 - 10. Notes / Assumptions**
 - Any clarifications, constraints, or assumptions that help interpret the use case.
 - Environment needs to be harsh enough to cause local network issues
 - Responders don't have training in networking so deployment must be simple
 - Nodes have a maximum of 25m spacing between one another
 - Since there is no power infrastructure so batteries lasts 15-20 minutes

Product Requirements

System Requirements

The Resilient Mesh Network Testbed consists of physical mobile nodes, virtual/simulated mesh nodes, an out-of-band test control and telemetry collection network, and a remote visualization dashboard [1], [2]. The requirements below establish operational, functional, environmental, and performance specifications of the system [1], [2]. This can be seen in our Requirements Matrix below.

Category	REQ-#	Priority	Requirement	Version #	Notes	Verification Type	Customer & VT Acceptance	Discussion
Functional	FUN-1	H	The system shall form an ad-hoc, multi-hop mesh network among its nodes with no access point, wired backhaul, or central router in the data path.	v1	Project request form §1.0.	DEMONSTRATION	Heil	Core premise of the project. The out-of-band test-control path (FUN-12) is not part of the data path.
	FUN-2	H	The system shall support communication between at least 4 active, moving physical nodes.	v1	Customer specified 4 moving nodes in 09/15 meeting.	TEST	Pending	Establishes minimum physical node capability of testbed.
	FUN-3	H	The system shall automatically integrate a node joining the network during operation without	v1	Customer named network join rate as a key dynamic (09/15 meeting).	TEST	Pending	Allows team to evaluate network behavior as topology changes.

			reconfiguring the other nodes.					
	FUN-4	H	The system shall automatically reroute traffic around a lost node or degraded link without operator intervention.	v1	Node loss will be simulated by powering down or removing node.	TEST	Pending	Required to evaluate network resiliency.
	FUN-5	H	The system shall carry a continuous data stream over UDP between designated nodes for the full test run, at rates configurable from low-rate telemetry to loads above network capacity, and record its delivery and continuity during topology changes.	v1	Customer strongly encouraged UDP (09/15 meeting). Request form specifies simulated sensor telemetry.	TEST	Pending	Demonstrates whether useful communication continues during failures and mobility. The stream is measured, not guaranteed: the customer counts a real protocol failure as a successful result. TCP retransmission would mask routing behavior.
	FUN-6	H	The system shall operate simulated nodes in the same mesh as the physical nodes, with physical nodes	v1	Customer wants scalability of 10+ nodes: 4 physical plus up to 6 simulated (09/15 meeting).	DEMONSTRATION	Pending	Allows us to evaluate larger swarm configurations without requiring additional physical hardware.

			routing traffic to and through simulated nodes.					
	FUN-7	H	The system shall place simulated nodes in the same coordinate frame as physical nodes, with simulated link behavior consistent with relative position.	v1	Customer asked to bridge the gap between physical and virtual nodes (09/15 meeting).	TEST	Pending	Keeps simulated nodes from behaving as if they were always in range.
	FUN-8	H	The system shall allow an operator to induce node loss, node join, and link degradation on physical and simulated nodes, and shall record the time of each induced event.	v1	Request form §1.0, §3.0. Customer approved attenuation / reduced TX power to degrade links (09/15 meeting).	DEMONSTRATION	Pending	The recorded event time is the start point for reconvergence measurement.
	FUN-9	H	The system shall allow each physical node to be moved along a planned path during a test run.	v1	Request form allows carts, rovers, or tracks; customer described robotics-style carriers (09/15). Propulsion method TBD.	DEMONSTRATION	Pending	Provides physical mobility for topology changes.

	FUN-10	H	The mesh routing should operate at OSI Layer 2.	v1	Customer leans toward Layer 2 (09/15 meeting).	INSPECTION	Pending	SHOULD, not SHALL: customer stated a preference, not a constraint.
	FUN-11	H	The system shall collect per-node routing state, per-link quality, and traffic statistics during each run, and shall compute reconvergence time after each induced node loss.	v1	Request form §1.0 lists reconvergence time, throughput/latency, and scaling as required metrics.	TEST	Pending	Source data for all resilience metrics in the final report.
	FUN-12	H	Test control and data collection shall not depend on the mesh network under test and shall keep working when it partitions.	v1	Team-derived from customer's validation requirement (09/15 meeting).	TEST	Pending	If data collection rides the mesh, it is lost exactly during the failure being measured.
	FUN-13	H	The system shall provide a means to determine whether an observed network failure is caused by the protocol or scenario	v1	Customer: failures must be real protocol failures, not our own errors (09/15 meeting). Customer's accepted evidence TBD.	ANALYSIS	Pending	Makes a demonstrated protocol failure defensible.

			under test rather than by a hardware, configuration, or implementation fault.					
	FUN-14	H	The system shall display live network topology, per-link quality, and node positions, distinguishing physical from simulated nodes.	v1	Request form §1.0 (visualization tool). GPS or relative coordinates acceptable (09/15 meeting).	DEMONSTRATION	Pending	Lets observers see topology change as it happens.
	FUN-15	H	The dashboard shall be viewable by the remote customer.	v1	Customer is in Idaho; public server ideal, local Python app plus recorded video acceptable (09/15 meeting).	DEMONSTRATION	Pending	Minimum is recorded video; live remote access is the goal.
Performance	PER-1	H	The system shall support at least 10 concurrent nodes (physical plus simulated) in one mesh.	v1	Customer: 4 physical plus up to 6 simulated; more than 10 conceptually (09/15 meeting).	TEST	Pending	Scale of the swarm scenario.
	PER-2	H	The system shall sustain an end-to-end application data rate of at least 2 Mbps across	v1	Customer asked for a few Mbps, enough for SD video (09/15). 2 Mbps is team-selected; to be confirmed by bench test.	TEST	Pending	Provides a measurable traffic load for evaluating network performance.

		a path of at least 2 hops.					
PER-3	M	The system shall reproduce single-hop link loss representative of 10–25 m node spacing within the available test area.	v1	Customer identified 10–15 m as typical spacing and 25 m as a test distance; attenuation or reduced TX power permitted (09/15).	TEST	Pending	Wi-Fi links hold well beyond 10 m, so link loss must be forced to create multi-hop topologies in a small area.
PER-4	H	With no induced stress, the system shall restore end-to-end delivery within 10 s after loss of a single relay node.	v1	Team-derived baseline. Value TBR after bench test.	TEST	Pending	Sanity check on the known-good baseline only. Reconvergence under stress is measured, not required.
PER-5	M	With no induced faults, the data stream packet delivery ratio shall be at least 95%.	v1	Team-derived baseline.	TEST	Pending	A baseline that already drops packets cannot validate a failure.
PER-6	H	During mobility runs, the system shall produce at least 1 topology change (link established or lost) per minute.	v1	Customer: drones at 60–200 mph, but dynamics matter more than speed (09/15). Value TBR pending scaling analysis.	ANALYSIS	Pending	Scales swarm speed by rate of topology change rather than vehicle speed.
PER-7	M	Simulated nodes shall be	v1	Customer named topology diversity as a	TEST	Pending	A single gateway would put

			attachable to the mesh through at least 2 different physical nodes.		key dynamic (09/15).			every simulated node behind one physical node.
	PER-8	M	Each physical node shall report its position with error no greater than 1.5 m at least once per second.	v1	Customer: GPS map or relative coordinates (09/15). Value TBR; GPS is unreliable indoors.	TEST	Pending	Position error must be small relative to link range.
	PER-9	H	The system shall measure reconvergence time with a resolution of 100 ms or better.	v1	Request form §1.0 (reconvergence time).	ANALYSIS	Pending	Requires a common time reference across nodes.
	PER-10	M	The dashboard shall reflect a topology change within 2 s of its occurrence.	v1	Request form §1.0 ('live' visualization).	TEST	Pending	Defines 'live'.
Security	SEC-1	L	The system should allow data-path encryption to be enabled as a configurable test variable.	v1	Customer: encryption is nice to have but not a focus; treat as a throughput factor (09/15).	TEST	Pending	Enables measuring the throughput cost of encryption.
	SEC-2	H	Any internet-reachable dashboard shall be read-only, require authentication	v1	Team-derived from remote-access requirement (FUN-15). VT hosting policy TBD.	TEST	Pending	A public server connects the internet to the lab network.

			n and encrypted transport, and provide no path to test control or node access.					
Compliance	CO M-1	H	All radio transmitters shall operate within FCC 47 CFR Part 15 limits in their certified configuration .	v1	Team-derived (US operation).	INSPECTION	Pending	Lowering power or adding attenuation is compliant; raising power or using higher-gain antennas than certified is not.
	CO M-2	H	No project artifact shall contain proprietary, FOUO, export-controlled, or classified information.	v1	Request form §4.0, §6.3: no NDA; posters and notebooks are public.	INSPECTION	Pending	All project output is public.
	CO M-3	H	Lithium battery charging, handling, and storage shall follow Virginia Tech EHS laboratory safety guidance.	v1	Team-derived (lab safety). Specific EHS document TBD.	INSPECTION	Pending	
Reliability	REL -1	H	Each physical node shall return to full operation after abrupt power removal, surviving at	v1	Node loss is simulated by powering down (request form §3.0).	TEST	Pending	Abrupt power loss can corrupt the storage of flash-booted boards.

			least 20 consecutive power cuts without storage corruption.					
Power	PO W-1	H	A single battery shall power a complete physical node, including mobility, for at least 20 minutes.	v1	Customer: 15–20 minute sorties are plenty (09/15).	TEST	Pending	Covers the longest sortie; avoids end-of-run voltage sag.
	PO W-2	M	A node battery shall be exchangeable within 5 minutes without tools.	v1	Customer: battery swapping practically required (09/15).	DEMONSTRATION	Pending	
	PO W-3	H	Each physical node shall detect and log undervoltage (brownout) events with a timestamp.	v1	Team-derived; supports FUN-13.	TEST	Pending	Brownouts otherwise look like node loss or link degradation.
Configuration	CON-1	H	Routing and test parameters shall be settable per run without code changes, and each run's full configuration shall be saved with its data.	v1	Customer emphasized documentation and full records (09/15).	INSPECTION	Pending	Makes every result reproducible.

	CON-2	H	Software will use open source and freely available operating systems and programs.	v1	Supports COM-2 and COST-1.	INSPECTION	Pending	Lower cost and supports reproducibility. Such as, creating more nodes, porting the design to a different OS, rewrite to different programming language if need be. Some parts may not be open source, like firmware or binary blobs.
	CON-3	M	The backend system, web dashboard server configuration and mesh node configuration, should be portable and not rely on any hard dependencies.	v1	Team-derived baseline.	INSPECTION	Pending	

Target Specifications

The target specifications quantify performance constraints, establishing marginal baseline limits, and ideal design targets derived from the Idaho National Labs customer meeting and initial team testing assumptions [4], [5].

Table 01-5 Target Specifications

Req ID	Metric	Units	Marginal Value	Target (Ideal) Value	Traceability / Rationale
PER-1	Active Concurrent Nodes	Nodes	10 nodes (4 physical, 6 simulated)	≥ 12 nodes	Customer requirement for swarm scaling beyond physical testbed bounds [1], [2].
PER-2	End-to-end multi hop throughput	Mbps	≥ 1.0 Mbps	≥ 2.0 Mbps	Supports continuous video/telemetry stream across 2+ hops [1], [2].
PER-3	Induced multi-hop attenuation	dB/m	Forced loss at 10m indoor radius	Scaled attenuation matching 25 m outdoor environment	Customer requirement for ideal testbed conditions. Forces multi-hop routing within constrained lab environments, [1], [2], [11].
PER-4	Reconvergence Recovery Time	Seconds	$\leq 10.0s$	$\leq 2.0s$	Baseline self-healing recovery speed upon node failure [1], [2], [5].
PER-5	Unstressed Packet Delivery Ratio	%	$\geq 90\%$	$\geq 98\%$	Prevents underlying link drop errors from showing routing failures [1], [2].

PER-6	Dynamic Topology Change Rate	Events/min	≥ 0.5 changes/min	≥ 1.0 changes/min	Models swarm dynamics independent of physical vehicle velocity [1], [2].
PER-7	Simulated Gateway Attachment	Interfaces	1 gateway	≥ 2 gateways	Ensures topology diversity and avoids single point of failure bottlenecks [1], [2].
PER-8	Physical Node Localization Error	Meters	$\leq 2.5\text{m}$	$\leq 1.5\text{m}$	Ensure accurate spatial positioning mapping for link quality overlay [1], [2].
PER-9	Event Timestamp Resolution	Milliseconds	$\leq 250\text{ ms}$	$\leq 100\text{ ms}$	Enables precise delta calculation for protocol reconvergence timing [1], [2].
PER-10	Visualization Update Latency	Seconds	$\leq 5.0\text{ s}$	$\leq 2.0\text{ s}$	Provides near real time visualization feedback for remote operators [1], [2].
POW-1	Single Battery Runtime	Minutes	$\geq 15\text{ min}$	$\geq 20\text{ min}$	Covers standard test run duration under full compute load [1], [2].
POW-2	Battery Swap Turnaround Time	Minutes	$\leq 5\text{ min}$	$\leq 2\text{ min}$	Minimizes downtime between consecutive tests [1], [2].

COST-1	Total Development and Fabrication Cost	USD	≤ \$7,500	≤ \$3,500	Matches Idaho National Labs project funding and VT lab resource sharing [1], [10]
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Target Specification Assumptions:

1. Physical node spatial testing will be conducted primarily in indoor laboratory environments or controlled outdoor environments [1], [2]. This requires artificial RF attenuation to simulate 25-meter link drops within smaller physical footprints [1], [2], [11].
2. GPS data can be supplemented by relative positioning or ultra-wideband tracking indoors where satellite reception is not reliable [1], [2].

Table 01-6 Standards and Statutory Requirements

Req. #	Requirement	Source Document (e.g., standard, regulatory requirements)	Details
COM-1	Unlicensed ISM Band Radio Operation	FCC 47 CFR Part 15 [3]	All 2.4 GHz / 5 GHz Wi-Fi radio transmissions must adhere to EIRP limits and certified antenna configurations [1], [3].
COM-2	Public Dissemination and Open Data	VT MDE Policy and INL Request 1]	Prohibits inclusion of proprietary, FOUO, ITAR or classified data [1] All code, hardware, and documentation are public domain [1].
COM-3	Battery Storage and Handling Safety	Virginia Tech EHS Lab Safety Standards [4]	Defines safety requirements for storage, charging, and disposal of multi cell Lithium Polymer or ion batteries [4].
SEC-2	Public Web Application Safety	OWASP Top 10 and VT IT Security Policy [6]	Ensures remote accessible visualization dashboards use TLS, authentication, and read only administrative isolation [1], [2], [6].

Benchmarking Information

To establish design tradeoffs for the self-healing mesh architecture, open-source routing protocols and alternative implementations were evaluated against key project metrics [4], [5], [8].

Table 01-7 Benchmarking Information

Feature / Requirement	Target Requirement	Alternative #1: BATMAN-adv (OSI Layer 2)	Alternative #2: AODV/OLSR (OSI Layer 3)	Alternative #3: Custom UDP Flood Protocol
FUN-10 Layer 2 Mesh Routing	OSI Layer 2 (Data Link) [2]	OSI Layer 2. Virtualizes nodes as standard virtual ethernet switches [9].	OSI Layer 3. Requires IP subnetting, NAT, and OS routing table modifications [9].	OSI Layer 7. Application-level forwarding over standard sockets.
PER-4 Reconvergence speed	≤ 2.0 seconds [1], [2]	Fast (100ms – 500ms) by periodic originator messages and link sensing [9].	Moderate (1.0-5.0s) due to route request discovery timeouts [9].	Variable (0.5 - 10.0s) dependent on application polling frequency.
PER-2 Throughput and Overhead	≥ 2.0 Mbps across 2+ hops [1], [2]	High efficiency. Minimal packet encapsulation overhead and handles non-IP frame traffic easily [9].	Moderate efficiency. IP-in-IP encapsulation adds packet overhead [9].	Low efficiency. High redundant message flooding overhead under stress.
FUN-6 Virtual Node Integration	Seamless physical/virtual routing [1], [2]	Native. Virtual network interfaces slide directly into layer 2 mesh [1], [9].	Complex. Requires tap tunnels, IP bridge networks, and virtual router instances [1].	Moderate. Virtual nodes run as local software processes sending UDP frames [1].

FUN-12 Out of Band Isolation	Data path fully independent from telemetry [1], [2]	Supported. Layer 2 mesh runs on dedicated Wi-Fi interface. Telemetry uses secondary network [1], [2].	Supported. Uses dual network interfaces for separated IP routing tables [1], [2].	Supported. Managed at the software application socket level [1], [2].
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